

Danish Mohammed

(630)-923-5307 | [Email](#) | [Github](#) | [LinkedIn](#)

Education

Illinois Institute of Technology | *Chicago, IL*

- Bachelor of Science in Computer Science | *Expected Dec 2027*

Y-Combinator Startup School | *San Francisco, CA* | *Summer 2026*

Professional Experience

AI & Data Science Intern | *Aarete* / *Jun-Aug 2026* / *Chicago, IL*

- Architected an end-to-end agentic Statement of Work (SOW) generation pipeline in LangGraph, orchestrating Claude Sonnet/Haiku over AWS Bedrock Knowledge Bases and exposing it firm-wide as an MCP tool that ingests a prompt plus supporting client documents and retrieves relevant context automatically.
- Built caching and parallelized processing nodes to reduce pipeline latency by ~500s, alongside four parallel quality-check nodes validating distinct aspects of each generated document
- Established an evaluation loop using historical SOW intake documents and “golden standard” outputs as a train/test set, with an LLM-as-judge scoring harness to iteratively tune prompting, node logic, and rules

Software Engineer (AI & Backend) | *Kaplan Institute (Illinois Tech)* | *Feb 2026 – May 2026*

- Engineered backend API infrastructure using FastAPI and Pinecone vector embeddings, powering a semantic search engine to intelligently index and match 1000+ student profiles. .
- Designed scalable PostgreSQL database schemas using asynchronous SQLAlchemy, managing complex state relationships for real-time scheduling, and dynamic UI hydration. Implemented auth with cognito and boto3
- Architected and deployed an autonomous AI Interviewer utilizing AWS Nova Sonic and LiveKit agents, integrated with a custom UI for low-latency, real-time candidate evaluations.

Technical Projects

Shape of Feeling (SAE-Lit) | Mechanistic Interpretability Research | *March 2026 – June 2026* | [Writeup](#)

- Investigated whether guilt is encoded as geometric structure — rather than a flat, undifferentiated cluster — in Gemma-2-2B’s activation space, using sparse autoencoders (SAEs). Causally verified six guilt-associated SAE features via forced-activation steering. Iterated and documented through failures with other methods and data
- Applied a pseudo-likelihood Ising correction to feature co-activation data — correcting for confounds that raw correlation could not see — and uncovered a four-feature causal chain with one hub feature, revealing a distributed structure rather than a monolithic representation.

Tomea: Autonomous Research Engine | *Python, Modal, LangChain, Rich (TUI)* | *Fall 2025 – Feb 2026*

- Architected an agentic system that autonomously parses, implements, and benchmarks ML research papers (e.g., ROSE, GDPO, MOE) directly from ArXiv IDs, using different “harnesses” based on paper type.
- Engineered a "Synthesize-Execute-Heal" loop: an LLM agent generates PyTorch implementations while a heuristic "Healer" agent autonomously diagnoses and fixes runtime errors (tensor mismatch, CUDA conflicts).
- Orchestrated serverless GPU training jobs on Modal to run parallel benchmarks, aggregating real-time training metrics into a custom Terminal User Interface (TUI) dashboard. Open Sourced on GitHub.

Multi-Agent RL Trading System (MARLTS) | *Python, PyTorch, NumPy* | *Fall 2025*

- Implemented Proximal Policy Optimization (PPO) from scratch, hand-deriving policy gradient mathematics to ensure robust understanding of agent learning dynamics. (Full notes and code on GitHub)
- Trained across 8 years of data, achieving 4–6% Sharpe ratio improvement when back testing on held-out data.

Skills

- **Languages:** Python, JavaScript/TypeScript, SQL, C++, C
- **Machine Learning:** PyTorch, Hugging Face Transformers, LangChain, Llama Index.
- **Infrastructure:** AWS (ECS, ECR, EC2, SageMaker), Docker, Modal, Bedrock, MCP, PostgreSQL

Leadership and Awards

Hackathon Winner (Overall & Best Use of Gemini) | *Illinois Institute of Technology* | *Nov 2025*

- Won "Overall Winner" by developing "Ace Interview," an AI-mock interview application, within 48 hours.

Founder/Engineer | *Startup Studio* | *Illinois Institute of Technology* | *Oct 2025 – Feb 2026*

- Advanced to the final round of the university’s startup accelerator

- Developed Tomea from a research project into a scalable product architecture under mentorship from IIT's venture head, gathering technical validation from engineers across major AI research labs